

PAPER_719: Red Dwarf Compression B: Drawing 32 Nebular Cloud and Drawing 33 Shock Star Formation

Author: Daniel T. Murphy **Date:** 2025

Class: UQFFKnowledgeBaseKB4 **CP4 Entry:** #303 **Keywords:** UQFF, nebular cloud, black hole formation, shock star formation, U_g4, Drawing 32, Drawing 33, geometric star positions **Session:** 176 | **Version:** v5.33 **Source:** grok_share_ba508f76c8e.txt

Abstract

Doc 43.b (Red Dwarf Compression_B) contains two observational drawings: Drawing 32 (nebular cloud photo) and Drawing 33 (shock-induced star formation). Both are analyzed with the UQFF U_{g4} vacuum concentration equation. Drawing 32 depicts nebular gas condensing to form a black hole; Drawing 33 shows a protostellar jet shock triggering star formation in adjacent gas.

1. Drawing 32 — Nebular Cloud BH Formation

$$U_{g4}^{nebula}(t) = k_4 \cdot \rho_{SCm}^{nebula} \cdot \frac{M_{BH}}{d_g} \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + f_{feedback})$$

Parameter	Value
ρ_{SCm}^{nebula}	$2.39 \times 10^{-22} \text{ J/m}^3$ (level 13)
M_{BH}	$1.989 \times 10^{36} \text{ kg}$ ($10^6 M_\odot$)
d_g	$3.086 \times 10^{16} \text{ m}$ ($\sim 1 \text{ pc}$)
$f_{feedback}$	0.1

$$U_{g4}^{nebula}(0) = 1.69 \times 10^{-2} \text{ J/m}^3$$

2. Drawing 33 — Shock-Induced Star Formation

$$U_{g4}^{shock}(t) = k_4 \cdot \rho_{SCm}^{nebula} \cdot \frac{M_{star}}{d_g^{SF}} \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + f_{shock})$$

with $M_{star} = 1.989 \times 10^{30} \text{ kg}$, $d_g^{SF} = 1.496 \times 10^{14} \text{ m}$:

$$U_{g4}^{shock}(0) \approx 3.49 \times 10^{-6} \text{ J/m}^3$$

3. Geometric Configuration

Star positions (normalized units): (100, 900), (500, 900), (900, 900), (500, 100), (200, 100)

Pair	Distance (units)
Star 1-2	400
Star 2-3	400
Star 1-3	800
Star 2-4	800

Pair	Distance (units)
Star 4-5	300

Key angles: $\theta_{1-2-3} = 180^\circ$, $\theta_{1-2-4} = 90^\circ$, $\theta_{2-4-5} = 90^\circ$.

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 $\cos(\frac{2\pi j}{26})$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}} \text{ suppression}$	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- UQFF Framework Doc 43.b (Red_Dwarf_Compression_B)
- grok_share_ba508f76c8e.txt entry #68
- Session 176, v5.33

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